Daily Logic Drill #5

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Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Spot the Flaw · Common Fallacies · Proof Critique · Valid vs. Invalid Arguments.

Section A

1. A student solves $\sqrt{x-2} = -3$ by squaring both sides to obtain $x - 2 = 9 \implies x = 11$. Does $x = 11$ satisfy the original equation?

Ans: False.

2. A student simplifies $\dfrac{x^2-4}{x-2} = 4 \implies x + 2 = 4 \implies x = 2$. Is $x = 2$ a valid solution to the original equation?

Ans: False.

3. In an attempted proof that $1 = 2$, a student starts with $a = b$, obtains $(a-b)(a+b) = b(a-b)$, and concludes $a + b = b$. Which algebraic operation is invalid?

Ans: Dividing by $a - b$ (division by zero is invalid).

4. A student claims that for all real x, $x^2 > 4 \implies x > 2$. Give a negative integer counterexample to this deduction.

Ans: -3

5. A student solves $\log(x) + \log(x-3) = 1$ and finds candidate solutions $x = -2$ and $x = 5$. Which candidate is extraneous?

Ans: -2

6. A student conjectures: "If $f'(c) = 0$, then f has a local maximum or minimum at $x = c$." Give a polynomial function $f(x)$ with $c = 0$ that disproves this claim.

Ans: $f(x) = x^3$ (an inflection point, no local extremum).

7. A student asserts: "$|a| < |b| \implies a < b$." Provide a counterexample pair of integers (a, b).

Ans: $(1, -2)$

8. A student concludes that because n^2 is divisible by 4, n must be divisible by 4. Find the smallest positive integer counterexample n.

Ans: 2

9. A student claims: "For all real numbers a, b, if $a < b$ then $a^2 < b^2$." Give a pair of integers (a, b) with $a < b$ and $a^2 > b^2$.

Ans: $(-5, 2)$

10. A student asserts: "If $A \subseteq B \cup C$, then $A \subseteq B$ or $A \subseteq C$." Give a set A of two elements and sets $B = \{1\}, C = \{2\}$ that disprove this claim.

Ans: $A = \{1, 2\}$

Section B

1. A student attempts to prove: "For all integers n, if n is odd, then n^2 is odd."
Line 1: Suppose n is odd, so $n = 2k + 1$ for some integer k.
Line 2: Then $n^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$.
Line 3: Since $2k^2 + 2k$ is an integer, n^2 is odd.
Is this proof valid? Justify your answer.

Ans: Yes, fully valid.

2. Find the flaw: "Claim: If $x^2 = 3x$, then $x = 3$." Proof: Divide both sides of $x^2 = 3x$ by x to get $x = 3$.

Ans: Dividing by x is invalid when $x = 0$.

3. Find the flaw: "Claim: The only real solution to $\sqrt{x+7} = x - 5$ is $x = 9$." Proof: Squaring both sides gives $x + 7 = x^2 - 10x + 25$, so $x^2 - 11x + 18 = 0$, so $(x - 9)(x - 2) = 0$, giving $x = 9$ or $x = 2$.

Ans: $x = 2$ is an extraneous root introduced by squaring; it must be checked against and rejected from the original equation.

4. A student argues: For every integer n, $6\left(\frac{4n+1}{2}-\frac{n-3}{2}\right)$ is a multiple of 3. *(after TMUA 2020 Paper 2 Q3)*
Line 1: $6\left(\frac{4n+1}{2}-\frac{n-3}{2}\right)=3\left(2\left(\frac{4n+1}{2}-\frac{n-3}{2}\right)\right)$.
Line 2: $=3(4n+1-n-3)$.
Line 3: $=3(3n-2)$.
Line 4: which is always a multiple of 3.
On which line does the argument first contain an error, and what exactly is that error?

Ans: Line 2

5. Find the flaw: "Claim: For every integer n, n^2+n+2 is even." Proof: If n is even, $n=2k$, so $n^2+n+2=4k^2+2k+2=2(2k^2+k+1)$, which is even. Hence n^2+n+2 is even for every integer n.

Ans: The proof only checks the even case; the odd case is never addressed.

6. Find the flaw: "Claim: For real x, $x^2=9 \iff x=3$." Proof: If $x=3$, then $x^2=9$. This proves the biconditional.

Ans: Only the $\Longleftarrow$ direction ($x=3 \implies x^2=9$) was shown; the $\Longrightarrow$ direction ($x^2=9 \implies x=3$) is false, since $x=-3$ also satisfies $x^2=9$.

7. Find the flaw: "Claim: For every positive integer n, the n-th Fibonacci number F_n (with $F_1 = F_2 = 1$) is either equal to 1 or prime." Proof: $F_3 = 2$, $F_4 = 3$, $F_5 = 5$ — all prime. Since the pattern holds for $n = 3, 4, 5$, the claim is proved. What is the smallest counterexample?

Ans: Checking finitely many cases does not prove a universal claim; $F_6 = 8 = 2^3$ is the smallest counterexample.

8. A student wants to prove: "If n is not divisible by 3, then n^2 is not divisible by 3." Instead, they show: "If n is divisible by 3, then n^2 is divisible by 3." Explain why this does not prove the original claim.

Ans: The statement shown is neither the original claim nor its contrapositive — it proves nothing about the original.

9. Find the flaw: "Claim: For all integers a, b, if a and b are both even, then $a + b$ is even." Proof: Since $a + b$ is even, $a + b = 2m$ for some integer m. Since a, b are even, $a = 2j, b = 2k$, so $2j + 2k = 2m$, i.e. $j + k = m$, which is true since m is an integer. Hence $a + b$ is even.

Ans: Circular reasoning: the proof begins by assuming $a + b$ is even (the very thing to be proved) instead of deriving it.

10. A student attempts to solve $\log_2(x-1) + \log_2(x+1) = 3$:
Line 1: $\log_2((x-1)(x+1)) = 3$.
Line 2: $(x-1)(x+1) = 2^3 = 8$.
Line 3: $x^2 - 1 = 8 \implies x^2 = 9$.
Line 4: $x = 3$ or $x = -3$.
Line 5: Hence the solution set is $\{-3, 3\}$.
On which line does the first error occur, and what is the true solution set? *(after TMUA 2017 Paper 2 Q13)*

Ans: Line 5 (only $x = 3$ is valid in the domain).

Section C

1. Consider the claim: "For every positive integer n, $2n^2 + 2n + 1$ is prime." *(after TMUA 2022 Paper 2 Q7)*

Ans: D

2. Let P: "$x^2 - 5x + 6 = 0$" and Q: "$x = 2$". A student argues $Q \implies P$ is true and concludes $P \iff Q$.

Ans: B

3. A student attempts to prove: "For every integer n, $n^2 + n$ is divisible by 2", checking only the even case.

Ans: B

4. A student attempts to prove: "No three consecutive odd positive integers can all be prime." *(after TMUA 2023 Paper 2 Q4)*

Ans: C

5. A student "proves" $\forall n\,\exists m\,(m > n)$ and concludes $\exists M\,\forall n\,(M > n)$.

Ans: B

6. A student "confirms" the converse of "rhombus $\implies$ perpendicular diagonals" using only the example of a square.

Ans: B

7. A student disproves $(*)$: "For every pair of real numbers x, y, $|x+y| < |x| + |y|$" using the pair $x = -2, y = -5$. *(after TMUA 2020 Paper 2 Q8)*

Ans: D

8. A student "verifies" "four equal sides $\iff$ square" by checking only that every square has four equal sides.

Ans: B

Section D

1. A flawed attempt to prove: if $s^2 \geq 4p$ then there exist real u, v with $u + v = s, uv = p$. *(after TMUA 2021 Paper 2 Q11)*

Ans: B

2. A student claims "$n^2 - n + 11$ prime $\implies$ $n^2 + n + 11$ prime" for all positive n, verified for $n = 1, \ldots, 9$.

Ans: (a) Finite verification is not a universal proof. (b) $n = 10$.

3. Evaluate a contradiction-style proof that "ab even $\implies$ a even or b even".

Ans: D) I and II only

4. Evaluate the "every odd prime > 3 has the form $6k \pm 1$" argument.

Ans: B

5. Evaluate: "For all $n \geq 2$, $n^2 - 1$ is composite", proved via $n^2 - 1 = (n-1)(n+1)$.

Ans: (a) Line 3. (b) $n = 2$; and $n = 2$ is the unique exception.